RESIDUAL SEISMIC CAPACITY EVALUATION FOR REINFORCED CONCRETE BUILDINGS CONSIDERING EFFECTS OF CHARACTERISTICS OF STRUCTURE AND EARTHQUAKE RESPONSE SPECTRUM

構造特性と地震動応答スペクトル形状を考慮した被災 RC 造建物の残存耐震性能評価

姚　倩倩（性能制御システム学）
Qianqian YAO

Abstract:
構造特性と地震動応答スペクトル形状が応答スペクトル法における被災 RC 造建物の残存耐震性能に及ぼす影響を検討した。最初に，　建物の強度と塑性率をパラメータとした建物モデルを作成し，　構造特性が建物の残存耐震性能に及ぼす影響を評価した。次に，　地震動応答スペクトルの周期と傾きをパラメータとした地震動モデルを作成し，　建物の一質点系モデルにおいて地震動応答スペクトル形状が残存耐震性能に及ぼす影響を評価した；また地震動応答スペクトルに基づく性能評価に使うスペクトルの選択をまとめました。

Keywords ： *residual seismic capacity ratio, earthquake response spectrum, structural characteristic, strength・deformation・damping*
耐震性能残存率，　応答スペクトル，　構造特性，　強度・変形・減衰

1. Introduction

To restore an earthquake damaged community as quickly as possible, a well-prepared reconstruction plan is essential. When an earthquake strikes a community and destructive damage to buildings occurs, quick damage inspections are needed to identify which buildings are safe and which are not in the case of aftershocks. So some researchers in Japan have proposed some methods to evaluate the residual seismic capacity of reinforced concrete structures based on residual seismic capacity ratio *R-index*, by considering failure mechanism. Residual seismic capacity ratio *R-index*, is evaluated by considering collapse mechanism and seismic capacity reduction factor η. Fig. 1 shows a conceptual diagram of the lateral force-displacement curve and a definition of the η factors. The η factor is defined as the ratio of residual energy dissipation capacity, E_r, to original energy dissipation capacity, E_d+ E_r, and can be calculated by Eq. (1). However, definition of η doesn't consider the earthquake response and although there are degradations of strength, deformation and energy dissipation for damaged member, η factor is effected by them comprehensively. Thus η factor cannot be adapted for evaluation of seismic capacity by equivalent linearization method, time history response analysis and so on.

$$\eta = \frac{E_r}{E_r + E_d} \tag{1}$$

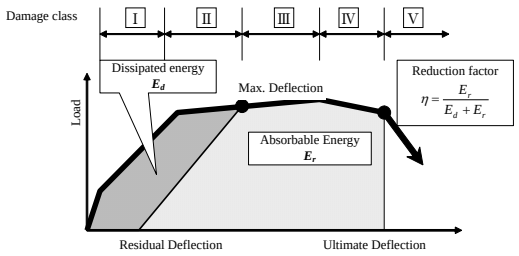

Fig.1 Seismic capacity reduction factor

Where, E_d: dissipated energy. E_r: residual energy dissipation capacity

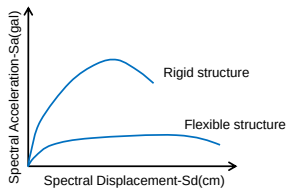

Fig.2 Flexible vs. rigid structure

On the other hand, since the Great Kanto earthquake in the year of 1923, the argument of flexible structure vs. Rigid structure has last for many years (see Fig.2). The purpose of this argument is to separate the periods of earthquake and building. The Japanese Standard for Seismic Evaluation of Existing Reinforced Concrete

Buildings (JBPDA) [1] proposed evaluation method of seismic capacity by using indexes of strength and ductility. However, even when both flexible structure and rigid structure are evaluated with same seismic capacity, they were still damaged in different levels in past earthquake. But the current evaluation method can't evaluate the effect of structure characteristics on residual seismic capacity.

Based on the above background, this thesis investigated residual seismic capacity of damaged RC buildings using Capacity Spectrum Method and evaluated the effects of characteristics of structure and earthquake response spectrum on residual seismic capacity.

2. Residual Seismic Capacity Evaluation using Capacity Spectrum Method

2.1 Introduction of Capacity Spectrum Method

The Capacity Spectrum Method (CSM), a performance-based seismic analysis technique, can be used for a variety of purposes. CSM method is used in Acceleration-Displacement Response Spectrum format (ADRS) (see Fig.3). The procedure compares the capacity of the structure (in the form a pushover curve) with the demands on the structure (in the form of response spectra). The graphical intersection of the two curves approximates the response of the structure. In order to account for non-linear inelastic behavior of the structural system, effective viscous damping values are applied to linear-elastic response spectra similar to inelastic response spectra. Effect viscous damping value F_h is calculated by Eq. (2).

$$F_h = 1.5/1+10h \tag{2}$$

h: damping factor

$$h = 0.05 + 0.25\left(1 - 1/\sqrt{\mu}\right) \tag{3}$$

μ: ductility factor

2.2 Seismic Capacity Index

In Japanese Guidelines for performance evaluation of earthquake resistant reinforced concrete buildings [2], Seismic Capacity Index α is defined as the ratio of capacity spectrum strength (S_{a1}) and demand spectrum strength (S_{a2}) as Eq. (4), shown in Fig.4.

$$\alpha = \frac{S_{a1}}{S_{a2}} \tag{4}$$

2.2 Residual Seismic Capacity Evaluation Method

In this thesis, residual seismic capacity index *R-index* is evaluated by the ratio between seismic capacity index of damaged building (α') and seismic capacity index of building with no damage(α),

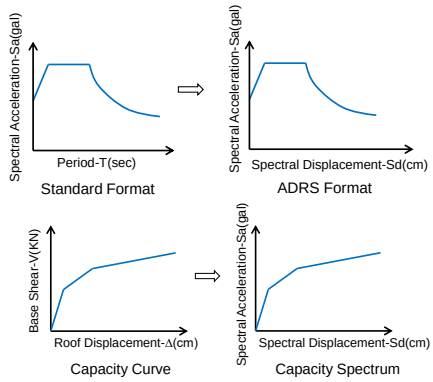

Fig.3 Capacity Spectrum Method

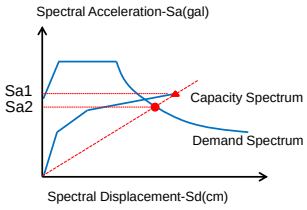

Fig.4 Seismic Capacity Index

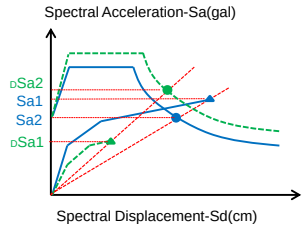

Fig.5 Residual Seismic Capacity

shown in Fig.5. The green line is capacity curve after building was damaged, and seismic capacity index of damaged buildings can be calculated by Eq. (5). So *R-index* can be evaluated by Eq. (6).

$$\alpha' = \frac{{}_D S_{a1}}{{}_D S_{a2}} \tag{5}$$

$$R = \frac{\alpha'}{\alpha} \tag{6}$$

3. Effect of Structure Characteristic on Residual Seismic Capacity

Based on standard damand spectrum in Japanese Guidelines [2] by considering the amplifying effect of the second type site which is often used, shown as Eq. (7), and Fig.6. Based on Hao's research[3], by using Capacity Spectrum Method, calculation of *R-index* is formulated by Eq. (8), (9). With regard to SDF model, performance reduction ratios of strength, deformation and damping are supposed to be equal with reduction ratios of structural elements shown inTable1, which were concluded in Yoshihiro ITO's research [4].

$$\begin{aligned} S_a &= 480 + 4500T\,(T \le 0.16) \\ S_a &= 1200\,(0.16 < T \le 0.864) \\ S_a &= 1036.8/T\,(0.864 < T) \end{aligned} \tag{7}$$

When T>0.864s

$$R = \sqrt{R_s} \cdot \frac{\sqrt{R_d} + 3R_h\sqrt{R_d} - 2.5R_h/\sqrt{\mu_0}}{4 - 2.5/\sqrt{\mu_0}} \tag{8}$$

When 0.16s<T<0.864s

$$R = \frac{R_s}{\sqrt{R_d}} \times \frac{\sqrt{R_d} + 3R_h\sqrt{R_d} - 2.5R_h/\sqrt{\mu_0}}{4 - 2.5/\sqrt{\mu_0}} \tag{9}$$

R_s: reduction ratio of strength

R_d: reduction ratio of deformation

R_h: reduction ratio of damping

μ_0: ultimate ductility of building

In order to evaluated effect of each performance reduction on residual seismic capacity, control variable method is used. *R-index* of each performance reduction is formulated as Eq. (10)~(15). The result is shown in Fig.7, 8. By defined performance reduction ratios in Table 1, Effect of ductility factor is shown as Fig.9.

When T>0.864s

$$R = \sqrt{R_s} \tag{10}$$

$$R = \frac{\alpha_d}{\alpha} = \frac{4\sqrt{R_d} - 2.5/\sqrt{\mu}}{4 - 2.5/\sqrt{\mu}} \tag{11}$$

$$R = \frac{\alpha_h}{\alpha} = \frac{1 + 3R_h - 2.5R_h/\sqrt{\mu}}{4 - 2.5/\sqrt{\mu}} \tag{12}$$

When 0.16s<T<0.864s

$$R = \sqrt{R_s} \tag{10}$$

$$R = \frac{1}{\sqrt{R_d}} \times \frac{4\sqrt{R_d} - 2.5/\sqrt{\mu_0}}{4 - 2.5/\sqrt{\mu_0}} \tag{13}$$

$$R = \frac{\alpha_h}{\alpha} = \frac{1 + 3R_h - 2.5R_h/\sqrt{\mu}}{4 - 2.5/\sqrt{\mu}} \tag{14}$$

3.3 Analyzing Result

With regard to each performance reduction, from Fig.7, When T>0.864s, R-index of deformation reduction is the lowest, so we can know that deformation reduction affect residual seismic capacity mostly; From Fig.8, When 0.16s<T<0.864s, damping reduction affect residual seismic capacity mostly.

With defined performance reduction ratios, from Fig.9, the effect tendency is too little, so that we also can ignore it.

4. Effect of earthquake response spectrum on Residual Seismic Capacity

4.1 Earthquake response spctrum characteristic model

In the study of GENG Shuwei [5], shape of earthquake response spectrum with period band longer than 0.864s is studied which is based on the ground motion records of west American and Taiwan Ji-Ji obtained by digital earthquake observing system. Based on these records with consideration of 5% damping, period within 6s, earthquake response spectrum S_a-S_d curve have been formulated into six types of F1~F6. S_a-S_d curve of F3 shown in Fig.10 is closest to the real response. Based on Eq. (7), (15), S_a-S_d curve is set into three

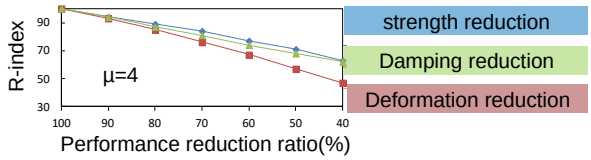

Fig.7 Effect of each performance reduction

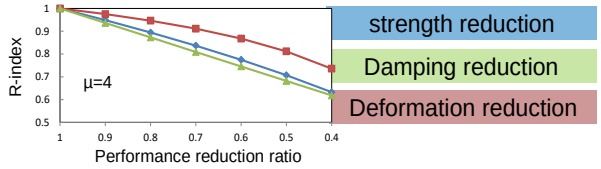

Fig.8 Effect of each performance reduction

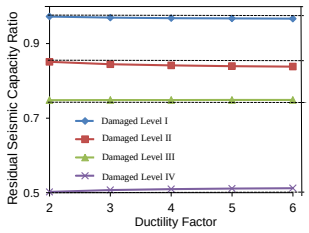

Fig.9 Effect of ductility factor

Table 1 Bending failure

Damage level	η	η_s	η_d	η_h
I (slight)	0.95	1.00	1.00	0.95
II (minor)	0.75	1.00	0.95	0.80
III (moderate)	0.5	1.00	0.85	0.75
IV (severe)	0.1	0.60	0.75	0.70

Shear failure

Damage level	η	η_s	η_d	η_h
I (slight)	0.95	1.00	1.00	0.90
II (minor)	0.6	1.00	0.85	0.70
III (moderate)	0.3	1.00	0.70	0.60
IV (severe)	0	0.40	0.50	0.50

stages of rising line, maximum strength and descent line which is shown in Fig.11. Formula of rising line and maximum strength is same with Eq. (7) of period T≤0.16, 0.16<T≤0.864. Based on Eq. (15), formula of descent line is set as Eq. (16).

Slope characteristic model (see Fig.12) and Period characteristic model (see Fig.13) were set.

$$F3 = a + (S_{a\max} - a)\left(\frac{T - 0.04}{T_1 - 0.04}\right)(0.04 \le T < T_1) \tag{15}$$

$$F3 = S_{a\max}(T_1 \le T < T_2)$$

$$F3 = S_{a\max}\left(\frac{T_2}{T}\right)^b (T_2 \le T < 6.0)$$

$$S_a = S_{a\max} \cdot (\beta_T / T)^\gamma \tag{16}$$

$$T = 2\pi\sqrt{\frac{S_d}{S_a}} \tag{17}$$

T : ultimate period

β_T : descent point period

S_{amax} : maximum strength value

γ : slope influence factor

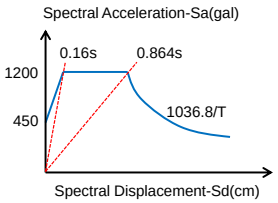

Fig.6 Standard earthquake response spectrum

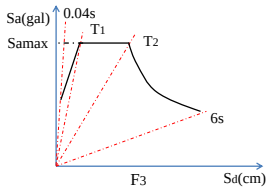

Fig.10 Spectrum of F3

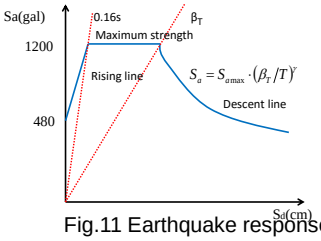

Fig.11 Earthquake response spectrum model

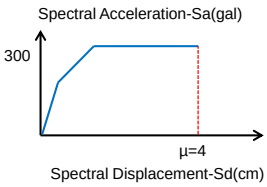

Fig.14 Building model

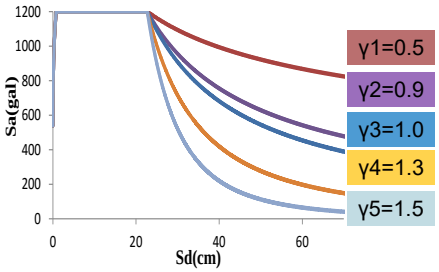

Fig.12 Slope factor model

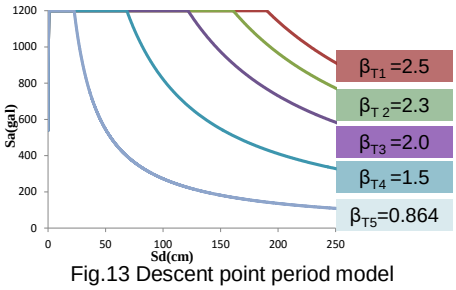

Fig.13 Descent point period model

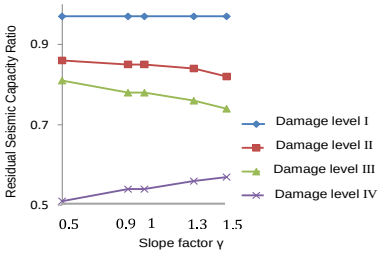

Fig.15 Effect of slope factor

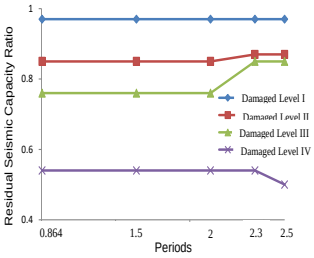

Fig.16 Effect of descent point period

4.2 Building Model

Analyzing building models is shown in Fig.14. Basis share coefficient is Q_y/W =0.3; maximum acceleration is 300gal; cracking yielding strength is one-third of maximum strength; yielding deformation angle is 0.5%; cracking deformation angle is 0.05%; ultimate ductility factor is 4.

4.3 Analyzing Result

Based on Capacity Spectrum Method, considering performance reduction ratio of strength (R_s), deformation (R_d) and damping (R_h), effect of slope characteristic is shown in Fig.15. Effect of period characteristic is shown in Fig.16. The performance reduction ratios are shown in Table 1. At damage level I, II, III, R-index becomes larger while slope factor γ becomes smaller; at damage level IV, R-index becomes smaller while γ becomes smaller; when decent point period becomes longer, it affect R-index: at damage level I, II, III, R-index becomes larger, at damage level IV, R-index becomes smaller.

5. Conclusion

1) Because evaluation of R-index by capacity spectrum method is result of ratio, so effect of strength can't be evaluated by CSM. The effect tendency of ductility factor is very little, so that we also can ignore it.

2) When buildings are slightly, minor, and moderately damaged, buildings have lower residual seismic capacity using earthquake response spectrum with large slope and small descent point period; severely damaged, buildings have lower residual seismic capacity using earthquake response spectrum with small slope factor and large descent point period. Based on the above conclusion, in order to evaluate residual seismic capacity conservatively with low R-index, when buildings are slightly, minor, moderately damaged, we'd better use spectrum with large slope factor and small descent point period; when buildings are severely damaged, we'd better use spectrum with small slope factor and large descent point period.

REFERENCE

[1] Japan Building Disaster Prevention Association (JBDPA): Standard for Seismic Evaluation of Existing RC Buildings, 2001.

[2] Architectural Institute of Japan (AIJ): Guidelines for Performance Evaluation of Earthquake Resistant Reinforced Concrete Buildings, 2004

[3] Linfei Hao, Qianqian Yao, Yoshihiro Ito, Yusuke Suzuki, Masaki Maeda, "Residual Seismic Capacity Assessment of Damaged RC Buildings Based on Response Spectrum Part 2. Assessment of residual seismic capacity ratio based on ideal model and comparison with seismic capacity reduction factor", published in the Architectural Institute of Japan, page 197-198, 2014.8.

[4] Yoshihiro ITO: Evaluation method of residualseismic capacity for damaged flames considering strength, deformation and energy dissipation, (Tohoku University), 2015.3

[5] GENG Shuwei: study on the provisions for the values of design response spectra in long period section, WORLD EATTHQUAKE ENGINEERING, Vol.24, No.2, Jun., 2008